Solstice Task 1

Mathematics Club

Instructions

- Provide descriptive answers for all questions. Explain your thought process well.
- This task if for pre-requisites. Section 1 and 2 are fairly straight- forward. Section 3 needs slight thinking. Section 4 onwards is expected to be heavy. Go step by step.
- Note on PODASIP questions: Prove or Disprove and Salvage if possible. So the question might be correct or incorrect. If it is correct, prove that it is. If it is wrong, prove that it is wrong. And if it was wrong and there exists a correct version, salvage.
- It is suggested and expected to not use LLMs (ChatGPT, Gemini, etc.) for thinking purposes. If the question asks you to derive/prove something, have fun struggling solving the problem and feeling the joy as you get closer to the answer. If you find yourself utterly stuck, do ask for hints on group or DM! One hour left before submission, hints did not work, go get some rescue from GPT. Although be mindful to not trust blindly and note that you are much smarter in thinking than GPT. It is very likely to hallucinate, specially PODASIPs ;)

All the best and have fun!

§1 What when you didn't just have an ex but also had y

(Sorry for the horrible joke). Remember how we discussed that our dataset of size n is sampled from a Random Variable $\{X_i\}_{i=1}^n$. But what happens when the data is not just X's but also Y's?

Let's switch to a concrete example: **Predicting Exam Scores based on Study Hours**.

- x: The number of hours a student studied.
- y: The marks obtained (0-100).

Consider a set of random variables (X^n, Y^n) which are independently drawn from a joint probability density function. If we assume they are independent and identically distributed (i.i.d.), the joint distribution is the product:

$$q(x_1, y_1)q(x_2, y_2) \cdots q(x_n, y_n)$$

We can factorize the distribution for a single pair as $q(x, y) = q(y|x)q(x)$.

Understanding the Distributions

1. The Input Distribution $q(x)$ (Study Habits):
This represents the probability of a student studying for x hours. Do we know the study habits of every student in the universe? No. But within our dataset, $q(x)$ tells us which study durations are common.

- $q(x = 4 \text{ hours})$ might be high (most students study a moderate amount).
- $q(x = 20 \text{ hours})$ is likely very low (almost no one studies 20 hours straight).

In our modeling context, we treat the input data x as "given" or "observed."

2. The Conditional Distribution $q(y|x)$ (The variability of Marks):
This is the "True Reality" we want to learn. It answers: *"Given that a student studied for 5 hours, what is the probability distribution of their score?"* Unlike a deterministic function where 5 hours $\rightarrow$ exactly 75 marks, real life has noise. If 100 students study for 5 hours, their marks will vary (some get 70, some 80, some 65). Thus, $q(y|x)$ is a distribution (likely a bell curve) centered around the "true" average score for that amount of studying.

The Triplet

So now our *model, truth, prior* triplet looks like $(p(y|x,w), q(y|x), \varphi(w))$:

- **The Model $p(y|x,w)$:** Our attempt to approximate reality. It predicts a score distribution y given hours x, tunable by weights w.
- **The Truth $q(y|x)$:** The actual, unknown distribution of marks in the real world that we are trying to learn.
- **The Prior $\varphi(w)$:** A known (we define(assume) it) probability distribution over the space W .

If you do not get a feel for this distribution view, drop a message in the group.

Let $X^n = (X_1, X_2, \ldots, X_n)$ be a set of random variables which are independently subject to a probability density function $q(x)$. For an arbitrary pair $(p(x|w), \varphi(w))$, the posterior probability density is defined by

$$p(w|X^n) = \frac{1}{Z(X^n)}\varphi(w)\prod_{i=1}^{n} p(X_i|w),$$

where $Z(X^n)$ is defined by

$$Z(X^n) = \int \varphi(w)\prod_{i=1}^{n} p(X_i|w)dw,$$

which is called the partition function or the marginal likelihood. The posterior average or the expected value over the posterior distribution is denoted by $\mathbb{E}_w[\ ]$. For an arbitrary function $f(w)$,

$$\begin{aligned}\mathbb{E}_w[f(w)] &= \int f(w)p(w|X^n)dw \\ &= \frac{1}{Z(X^n)}\int f(w)\varphi(w)\prod_{i=1}^{n} p(X_i|w)dw.\end{aligned}$$

The posterior variance is also defined by

$$\mathbb{V}_w[f(w)] = \mathbb{E}_w[f(w)^2] - \mathbb{E}_w[f(w)]^2.$$

As discussed, note that the expectation operator $\mathbb{E}_w[\ ]$ depends on a set X^n, hence $\mathbb{E}_w[f(w)]$ is not a constant but a random variable. The predictive density function is defined by

$$p(x|X^n) = \mathbb{E}_w[p(x|w)] = \int p(x|w)p(w|X^n)dw.$$

For a given sample X^n, the Bayesian estimation of the true distribution is defined by $p(x|X^n)$.

Write the Posterior distribution and marginal likelihood for this statistical model. Write Posterior Average and Posterior Variance. Write the Predictive Distribution. Write all of these as comparison from the previous scenario of a single random variable $\{X\}_{i=1}^{n}$.

§2 Gradient Descent v/s Bayesian Prediction

Given a dataset D_n of inputs $x \in \mathbb{R}^N$ and outputs $y \in \mathbb{R}^M$ drawn from some true distribution $q(y, x)$, the objective is to train a model $p(y \mid x, w)$ to predict an output. This amounts to estimating a parameter vector w that minimises a loss metric.

In the below question, whichever step is not done, fill in the blanks. It is just for generic understanding.

Comparison Example: Linear Regression

Let's compare the two approaches using a simple linear model: $y = mx + c$. Here, our parameter vector is $w = \{m, c\}$, where m is the slope and c is the intercept.

The Dataset (D_3): Three points perfectly lying on the line $y = 1x + 1$.

- $(x_1, y_1) = (0, 1)$
- $(x_2, y_2) = (1, 2)$
- $(x_3, y_3) = (2, 3)$

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1. Gradient Descent

Goal: Start at random coordinates and iteratively step towards the bottom of the error valley.

Step 1: Define Loss We use Sum of Squared Errors (SSE) with respect to parameters m and c:

$$L(m, c) = \sum_{i=1}^{3} (y_i - (mx_i + c))^2$$

Step 2: Calculate Gradients

Step 3: Perform One Iteration (Hand Calculation) Initialize $m = 0, c = 0$. Learning rate $\eta = 0.1$.

1. **Predict:**
   - $x = 0 \to \hat{y} = 0(0) + 0 = 0$ (Error: $1 - 0 = 1$)
   - $x = 1 \to \hat{y} = 0(1) + 0 = 0$ (Error: $2 - 0 = 2$)
   - $x = 2 \to \hat{y} = 0(2) + 0 = 0$ (Error: $3 - 0 = 3$)
2. **Compute Gradients:**
$$\frac{\partial L}{\partial m} = ...$$
$$\frac{\partial L}{\partial c} = ...$$
3. **Update Parameters:**
$$m_{new} = 0 - 0.1(...) = ...$$
$$c_{new} = 0 - 0.1(...) = ...$$

Result: After one step, we have specific coordinates ($m = ..., c = ...$). We repeat this loop until convergence.

Bonus: How to define convergence? How is it related to the learning step?

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2. The Bayesian Approach

Goal: Overlay our prior belief with the data constraints to create a 2D probability map.

Step 1: Define Prior Assume we believe m and c are small numbers, normally distributed.

$$P(m,c) \propto \exp\left(-\frac{m^2+c^2}{2}\right)$$

Step 2: Define Likelihood (The Data Constraints) Every data point adds a constraint. The likelihood is the product of the probabilities for all 3 points.

$$P(D|m,c) \propto \exp\left(-\frac{(1-(m(0)+c))^2}{2}\right) \cdot \exp\left(-\frac{(2-(m(1)+c))^2}{2}\right) \cdot \exp\left(-\frac{(3-(m(2)+c))^2}{2}\right)$$

Step 3: The Posterior (The Combined Map) To get the answer, we multiply Prior and Likelihood (sum the exponents):

$$P(m,c|D) \propto \exp\left(-\frac{1}{2}\left[\underbrace{m^2+c^2}_{\text{Prior}} + \underbrace{(1-c)^2+(2-m-c)^2+(3-2m-c)^2}_{\text{Likelihood}}\right]\right)$$

Result: Notice we don't calculate a "step." We have derived a **surface equation**.

- The expression inside the exponent describes a quadratic "bowl" in 2D space (m,c).
- The minimum of this bowl is the "Most Likely" solution.
- The *width* of the bowl tells us how uncertain we are about the slope m and intercept c.

Bonus: The way you took lattice for the probability distribution over μ, σ, do that for m, c.

Key Difference

- **Gradient Descent:** You are a hiker in a dark valley taking steps downhill to find the bottom. You only know where your feet are right now.
- **Bayesian:** You turn on the lights. You see the entire shape of the valley (the distribution). You know exactly where the bottom is and how steep the walls are.

§3 Estimations again

Let $w_0 = (1,1,1,\ldots,1) \in \mathbb{R}^{10}$ be the true parameter $(q(x) = p(x|w_0))$and W be a random variable on $\mathbb{R}^{10}$ which is subject to

$$p(w) = C\{\exp(-\|w\|^2) + 100\exp(-10\|w-w_0\|^2)\},$$

where $\|w\|^2 = \sum_i w_i^2$ and C is a constant.

PODASIP the following:

(1) Let $\hat{w}$ be the maximum point of $p(w)$. Then $\hat{w} \approx w_0$.

(2) $\mathbb{E}[W] \approx 0$.

Hence $\mathbb{E}[W]$ is far from $\hat{w}$. Assume that $p(w)$ is a posterior distribution of some statistical model(So it actually is some p(w|D), just notational difference). Discuss the difference between the maximum likelihood or *a posteriori* estimator and Bayesian estimation.

§4 Losses and Errors

§1 Entropy

The average and empirical entropies of the true distribution are respectively defined by

$$S = -\int q(x) \log q(x) dx, \tag{1.1}$$

$$S_n = -\frac{1}{n} \sum_{i=1}^{n} \log q(X_i). \tag{1.2}$$

Then by the definition,

$$\mathbb{E}[S_n] = S \tag{1}$$

$$\mathbb{V}[S_n] = \frac{1}{n} \left[\int q(x)(\log q(x))^2 dx - S^2 \right] \tag{2}$$

What do you think Entropy means for any probability distribution? Hint: "Surprisal"

Why do you think we take log?

§2 Generalization and Training Loss

Let X^n be a sample which is independently taken from a true distribution $q(x)$ and $p(x|X^n)$ be a predictive density using a statistical model $p(x|w)$ and a prior $\varphi(w)$. The training and generalization losses are respectively defined by

$$T_n = -\frac{1}{n} \sum_{i=1}^{n} \log p(X_i|X^n) \tag{3}$$

$$G_n = -\int q(x) \log p(x|X^n) dx \tag{4}$$

Note that both G_n and T_n are random variables. Let S be the entropy of a true distribution. Calculate $G_n - S$. When is $G_n - S = 0$? What is $G_n - S$ widely known as?

Refelct: Can Generalization loss be calculated in real world? Can Cross Validation be? What would cross validation depend on?

Bonus: Averages of C_n and G_{n-1} are equal but what about variance? What about variances of $G_n - S$ and $C_n - S_n$?

PODASIP: T_n is a better metric than G_n if I were to compare two models $(p_1(x|w), \varphi_1(w)), (p_2(x|w), \varphi_2(w))$ and their respective predictive densities $p_1(x|X^n), p_2(x|X^n)$.

Hint: Use Entropy

§3 Cross Validation Loss

Assume $n \geq 2$. Let $X^n \setminus X_i$ be a set of random variables $X_1, X_2, \ldots, X_n$ which does not contain X_i and $p(x|X^n \setminus X_i)$ be the predictive density using $X^n \setminus X_i$. The cross validation loss is defined by

$$C_n = -\frac{1}{n}\sum_{i=1}^{n} \log p(X_i|X^n \setminus X_i). \tag{5}$$

Also $C_n - S_n$ is called a cross validation error.

Reflect: Why do we call it error by subtracting the entropy of true distribution?
Hint: I want to compare learning across different true distributions

PODASIP:

Assume that X^n is independent. Then the following holds.
(1) Assume that the expectation values of G_n and C_n are finite. Then

$$\mathbb{E}[C_n] = \mathbb{E}[G_{n-1}]. \tag{6}$$

(2) The cross validation loss satisfies the relation,

$$C_n = -\frac{1}{n}\sum_{i=1}^{n} \log\left(\mathbb{E}_w[p(X_i|w)]\right).$$

(3) For an arbitrary set of random variables X^n,

$$C_n \geq T_n.$$

The equality $C_n = T_n$ holds if and only if $p(X_i|w)$ $(i = 1, 2, \ldots, n)$ is a constant function of w on $\{w \in W; p(w|X^n) > 0\}$.

Hint: For (3), look up Cauchy Schwartz inequality.

§5 Free ~~S Grade~~ Energy

We know that $\varphi(w)$ is a PDF over W. Prove that

$$\int Z(X^n) dx_1 dx_2 \cdots dx_n = 1$$

Where $Z(X^n) = \int \varphi(w)\Pi_{i=1}^{n} p(X_i|w)dw$ the denominator in Bayes rule for finding Posterior distribution.

Therefore $Z(x^n)$ can be understood as an estimated probability density function of Random Variable X^n obtained by using the statistical model $p(x|w)$ and $\varphi(w)$. $Z(x^n)$ is sometimes written as $p(x^n)$ (Read this paragraph thrice slowly steadily.)

The free energy or the negative log marginal likelihood is defined as $F_n = -\log Z(X^n)$. Then if the $q(x^n) = \Pi_{i=1}^{n} q(x_i)$, what is

$$\mathbb{E}[F_n] - nS$$

, where S is the entropy of the true distribution $q(x)$.

Reflect: What is the difference between $\mathbb{E}[G_n] - S$ and $\mathbb{E}[F_n] - nS$?

§6 Summing up

(Coin toss) Let $p(x|a)$ be a statistical model of $x \in \{0, 1\}$ which is defined by

$$p(x|a) = a^x (1-a)^{1-x},$$

where a is a parameter ($0 \leq a \leq 1$). See Figure below. Let us study a prior $\varphi(a)$ which is defined by

$$\varphi(a) = 1,$$

on $0 \leq a \leq 1$. Let X^n be a set of random variables which are independently subject to $p(x|a_0)$. Also let n_1 and n_2 be

$$n_1 = \sum_{i=1}^{n} X_i, \quad n_2 = n - n_1.$$

Then solve the following equations.

1. What is the maximum likelihood estimator is $\hat{a}$ and the estimated probability distribution $p(x|\hat{a})$ by the maximum likelihood method
2. PODASIP: The Bayesian predictive distribution $p(x|X^n)$ is

$$p(1|X^n) = \frac{n_1 + 1}{n + 2}, \quad p(0|X^n) = \frac{n_2 + 2}{n + 1}.$$

3. Derive that the generalization error of the maximum likelihood method as defined by Kullback-Leibler information from $p(x|a_0)$ to $p(x|\hat{a})$ is,

$$\begin{aligned} K_{ML} = &-a_0 \log((n_1 + \varepsilon)/(n + 2\varepsilon)) - (1 - a_0) \log((n_2 + \varepsilon)/(n + 2\varepsilon)) \\ &+ a_0 \log a_0 + (1 - a_0) \log(1 - a_0), \end{aligned}$$

   where $\varepsilon > 0$ is a sufficiently small positive value. Note that, if $\varepsilon = 0$, $0 < a_0 < 1$, and $n_1 n_2 = 0$, then $K_{ML} = \infty$. That of Bayesian method is defined by Kullback-Leibler distance from $p(x|a_0)$ to $p(x|X^n)$. What is it in terms of a_0, n_1, n_2, n?

4. By using numerical calculation, the expected values of $\mathbb{E}[K_{ML}]$ and $\mathbb{E}[K_{Bayes}]$ for $n = 20$, $a_0 = 0.05, 0.10, \ldots, 0.50$, and $\varepsilon = 0.0001$ are shown in Figure below. If $a_0 = 0$, then $\mathbb{E}[K_{ML}] = 0$ and $\mathbb{E}[K_{Bayes}] > 0$. Discuss the difference between the maximum likelihood and Bayesian methods from the viewpoint of the generalization errors.
   Bonus: Get a similar plot by coding up for $n = 30$

   PS: The ε is above problems is to avoid dividing by zero. Also Laplace's smoothing (adding 1) is used in problem 2 for similar reasons. Nothing fancy it is, all that is done is adding 1.

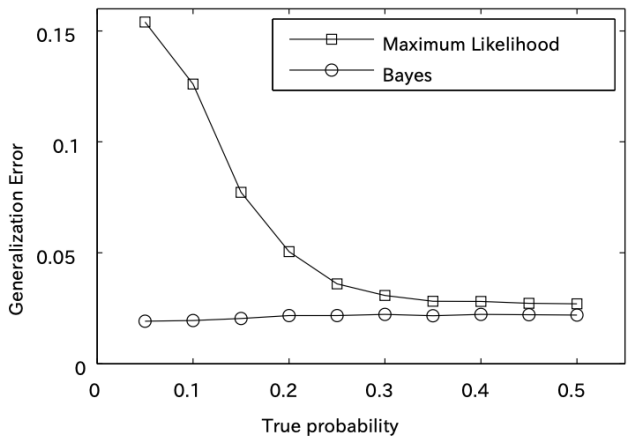

Figure 1.7: Comparison of Bayes with maximum likelihood in coin toss. The averages of the generalization errors by the maximum likelihood and Bayesian methods are compared in a coin toss problem for $n = 20$. The standard deviations of the maximum likelihood method are larger than that of Bayes for every a_0.

Source: Mathematical Theory of Bayesian Statistics by Sumio Watanabe